McMaster University

Capstone Project

Department of Computing and Software

Software Engineering and Management

Chemware Engineering

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Our Supervisor

Dr. Mahalec
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Motivation

Importance of Modeling & Simulation: Essential for optimizing efficiency, safety, and design for plants and factories.

Limitations of Existing Software:
- Rigid constraints on model structures, node interactions, and simulation types.
- Limited ability to create highly customized and complex process models.

Our Solution:
- Developed a web-based platform with unprecedented flexibility for process modeling.
- Users can define custom nodes and edges tailored to specific process requirements.
- Enables running specialized simulations suited to individual needs.

Key Benefits:
- Allows researchers, engineers, and students to explore a broader range of process scenarios.
- Overcomes the limitations of traditional modeling tools by offering greater configurability.

Collaboration & Goal:
- Developed in collaboration with McMaster University's Chemical Engineering department.
- Aims to provide a versatile and user-driven approach to process simulation.

Technologies Used

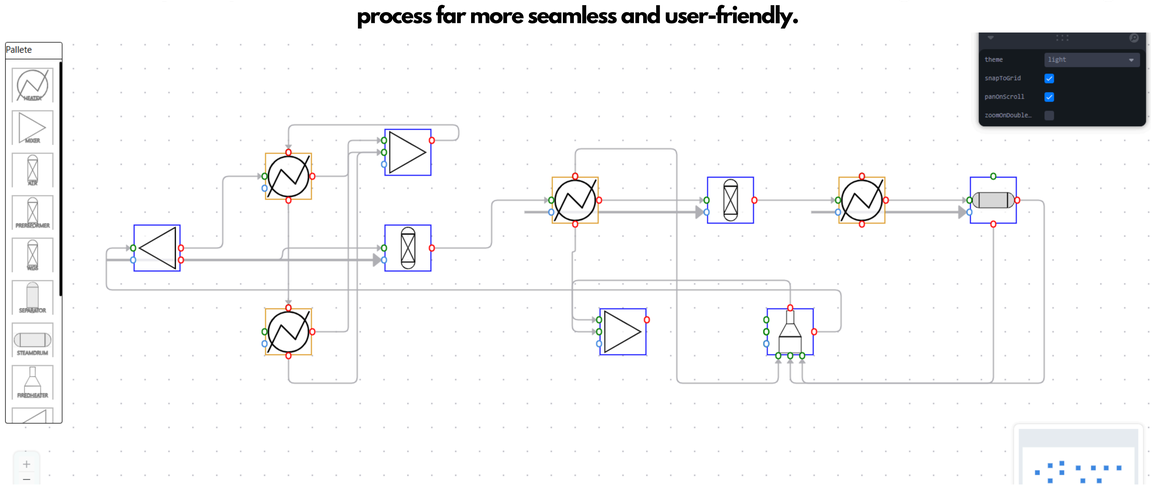
React Flow, Python, express, docker, mongoDB, PostgreSQL, TS, node JS, Prisma

Previous v/s Current

9	CM	CM*compname*	ME	component material				
10	CM	CM*compname*	ME0	component material				
11	CMS	CM*compname*	M	component material stream				
12	CMS	CM*compname*	ME	component material stream				
13	CMS	CM*compname*	ME0	component material stream				
14	E	E	E	energy				
15	E	E	ME	energy				
16	E	E	ME0	energy				
17	EF	EF	E	energy flow				
18	EF	EF	ME	energy flow				
19	EF	EF	ME0	energy flow				
20	EL	EL	E	electric energy				
21	EL	EL	ME	electric energy				
22	EL	EL	ME0	electric energy				
23	ELF	ELF	E	electric energy flow (power)				
24	ELF	ELF	ME	electric energy flow (power)				
25	ELF	ELF	ME0	electric energy flow (power)				
26	M	M	M	material				
27	MS	MF	M	material stream				
28	MS	MF	ME	material stream				
29	MS	MF	ME0	material stream				
30	MES	MF	ME0	material and energy stream		N		
31	MES	T0	ME0	material and energy stream	Y	N		
32	MES	P	ME0	material and energy stream	Y	N		
33	MES	H	ME0	material and energy stream	Y	N		

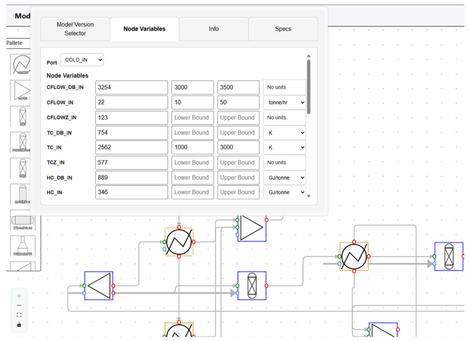
Previously, defining nodes and edges for plant models required users to manually input data into an Excel sheet. This process was not only time-consuming but also prone to human error, making it difficult to build and modify complex models efficiently.

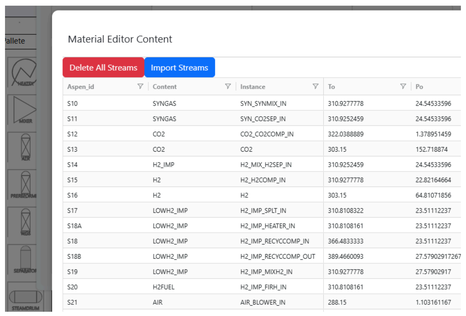
To address this, Dr. Mahalec proposed a more intuitive solution: a graphical user interface (GUI) that allows users to visually construct their models. Instead of representing a chemical plant as rows of data in a spreadsheet, users can now build and edit models using diagramming tools, significantly improving accuracy, usability, and efficiency. With this new approach, users can directly manipulate nodes and edges in an interactive environment before running simulations, making the process far more seamless and user-friendly.



Key Accomplishments

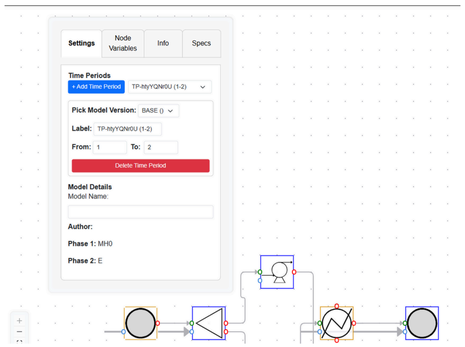
Canvas
- Add, update, and delete nodes
- Add node parameter data
- Complete actions from the canvas





Material Editor
- Read existing streams
- Delete old or unused streams
- Import new set of streams based on use case

Time Period
- Model verification system
- Interaction with various time periods
- Changing individual time period node data



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